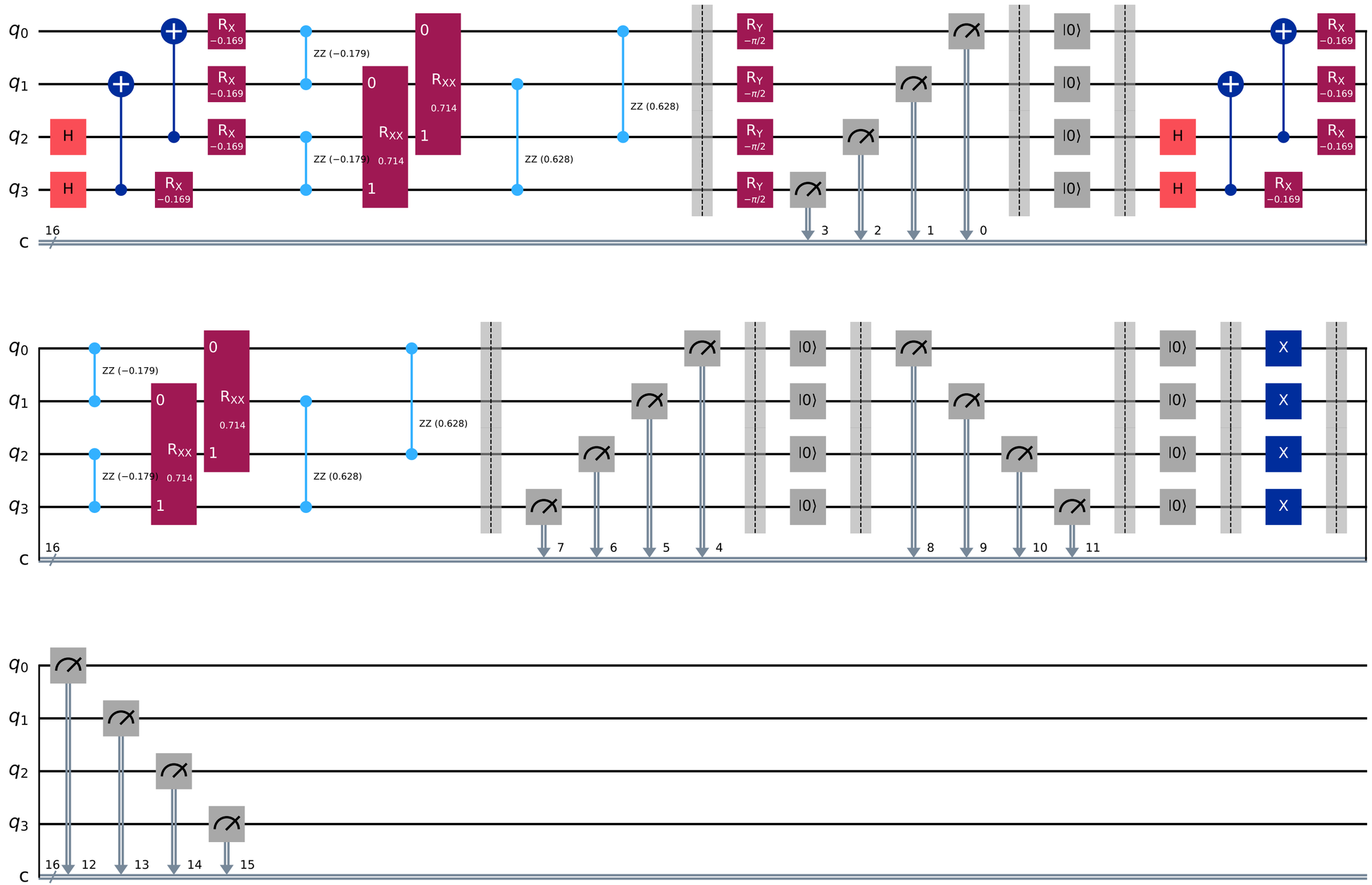
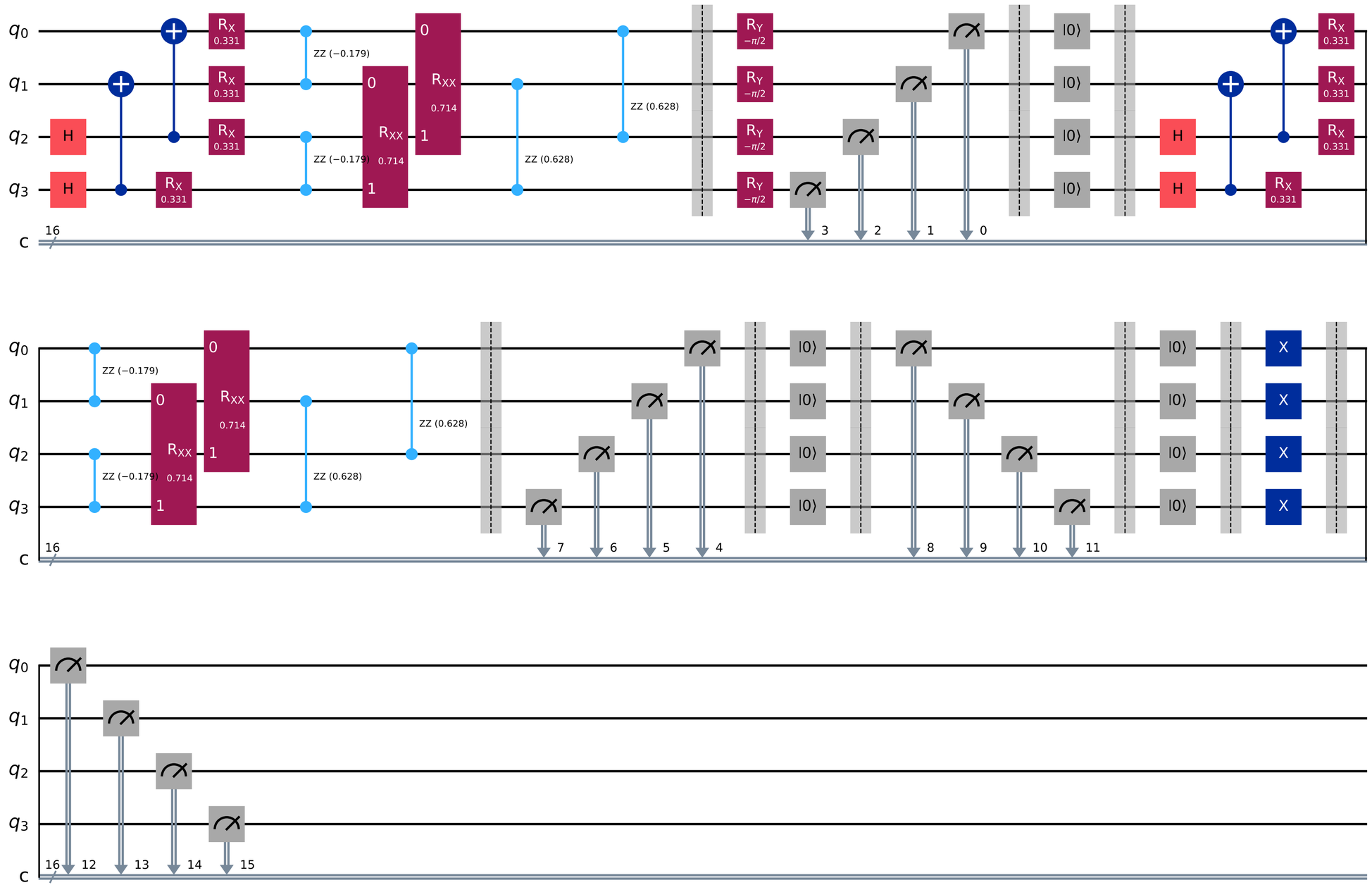
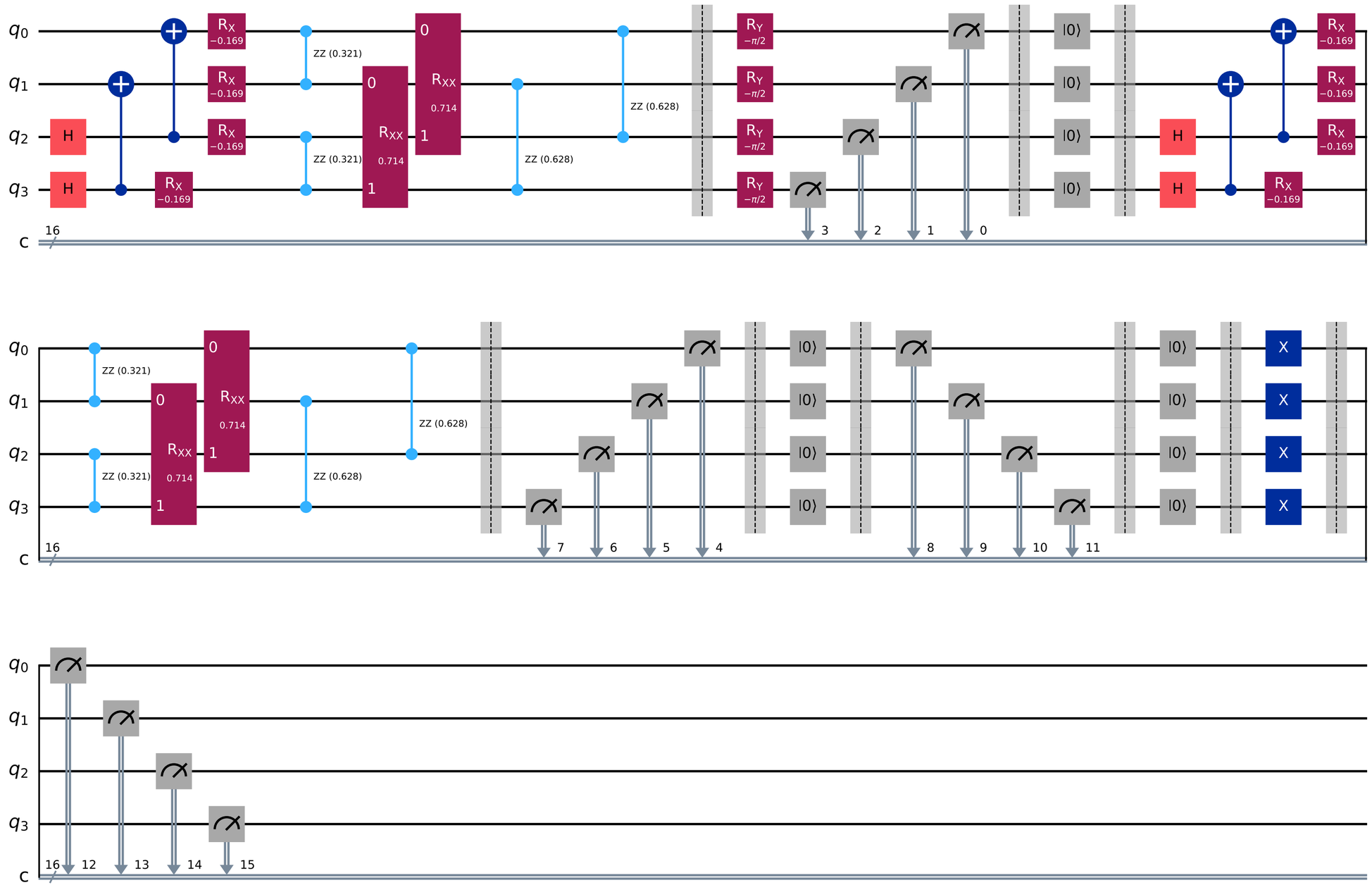
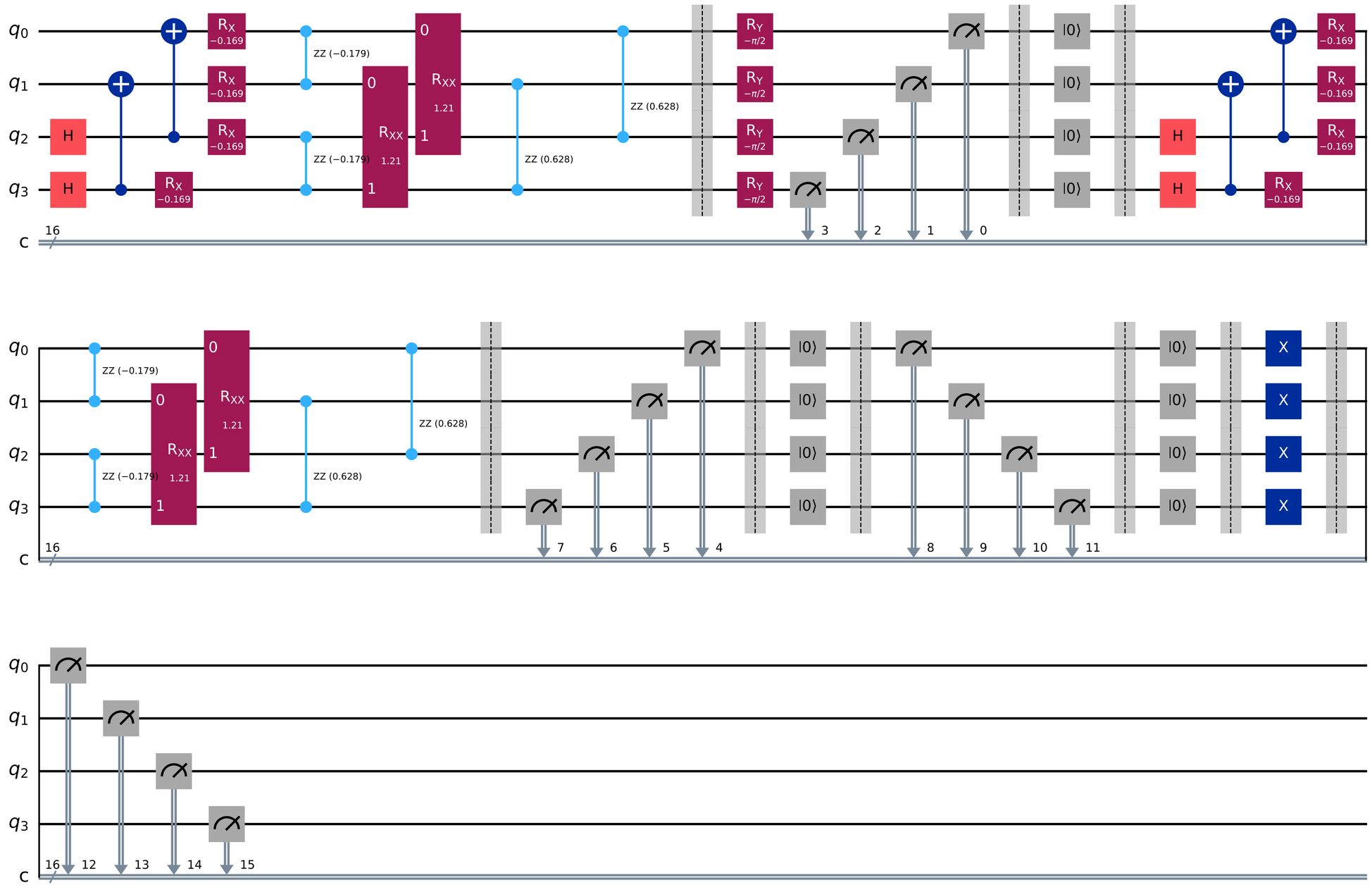
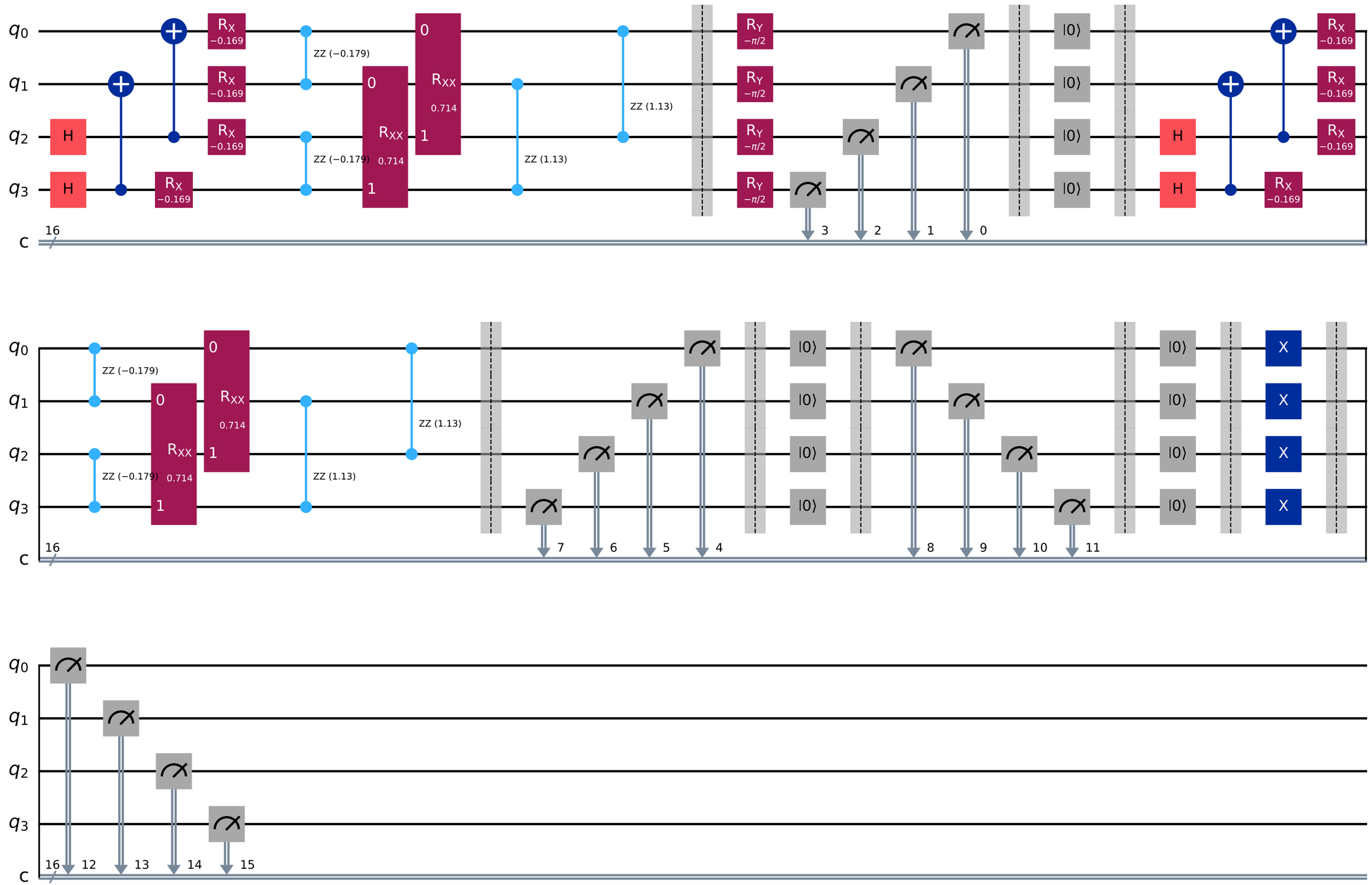
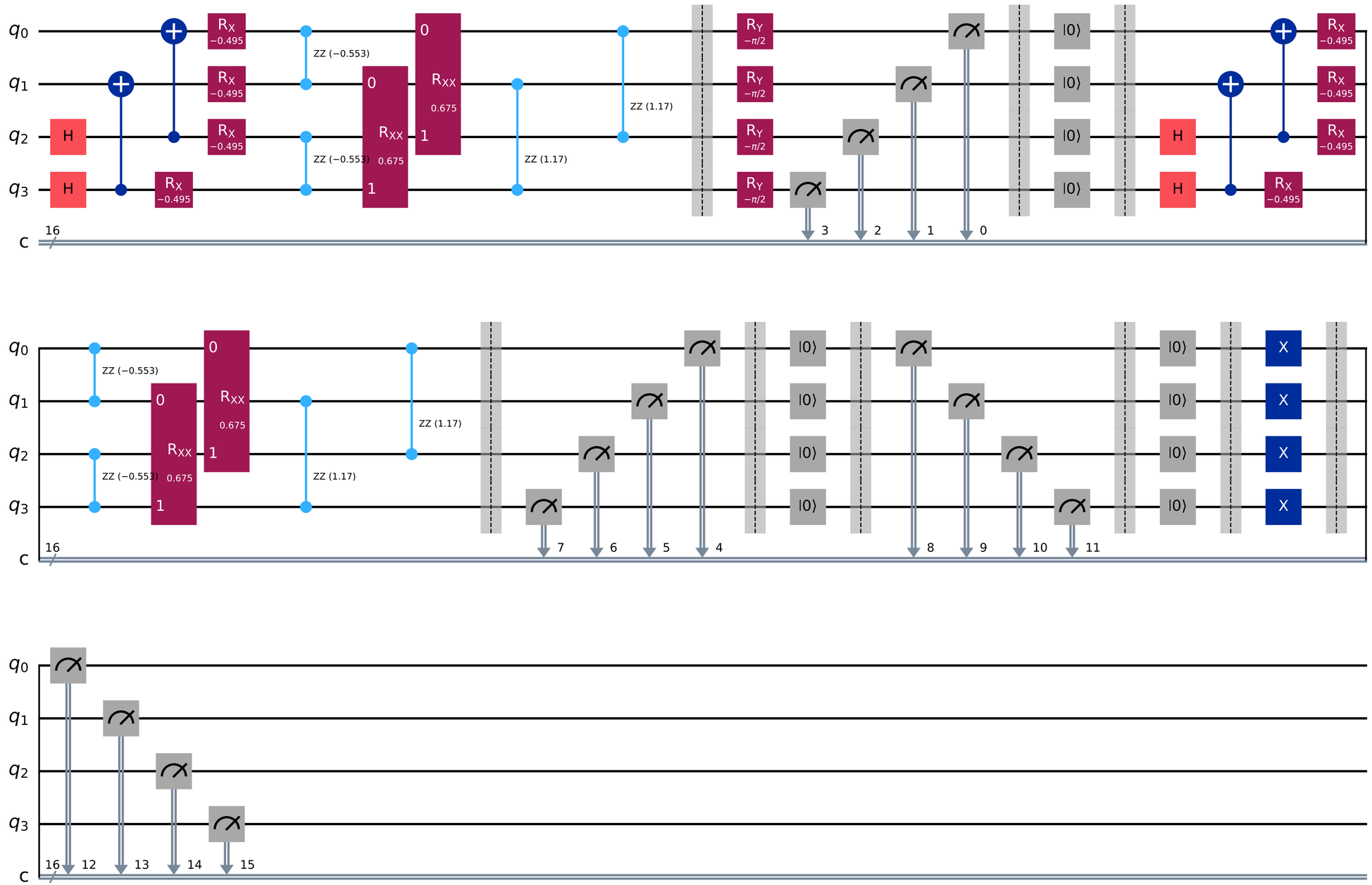

q0
q1
q2
q3
c
16
H
H
RX −0.169
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (0.628)
ZZ (0.628)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (0.628)
ZZ (0.628)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

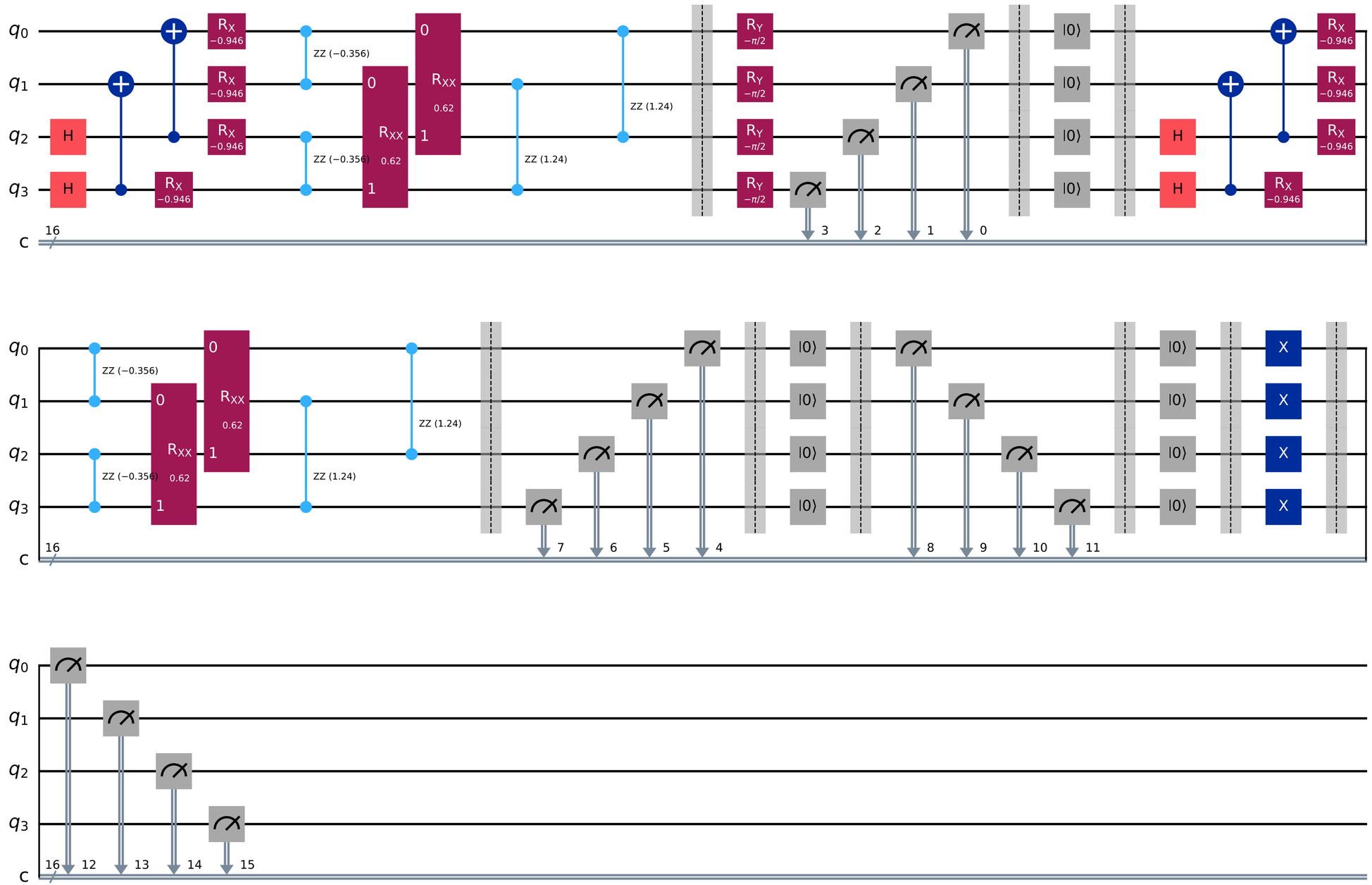

q0
q1
q2
q3
c
16
H
H
RX 0.331
RX 0.331
RX 0.331
RX 0.331
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (0.628)
ZZ (0.628)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.331
RX 0.331
RX 0.331
RX 0.331
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (0.628)
ZZ (0.628)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

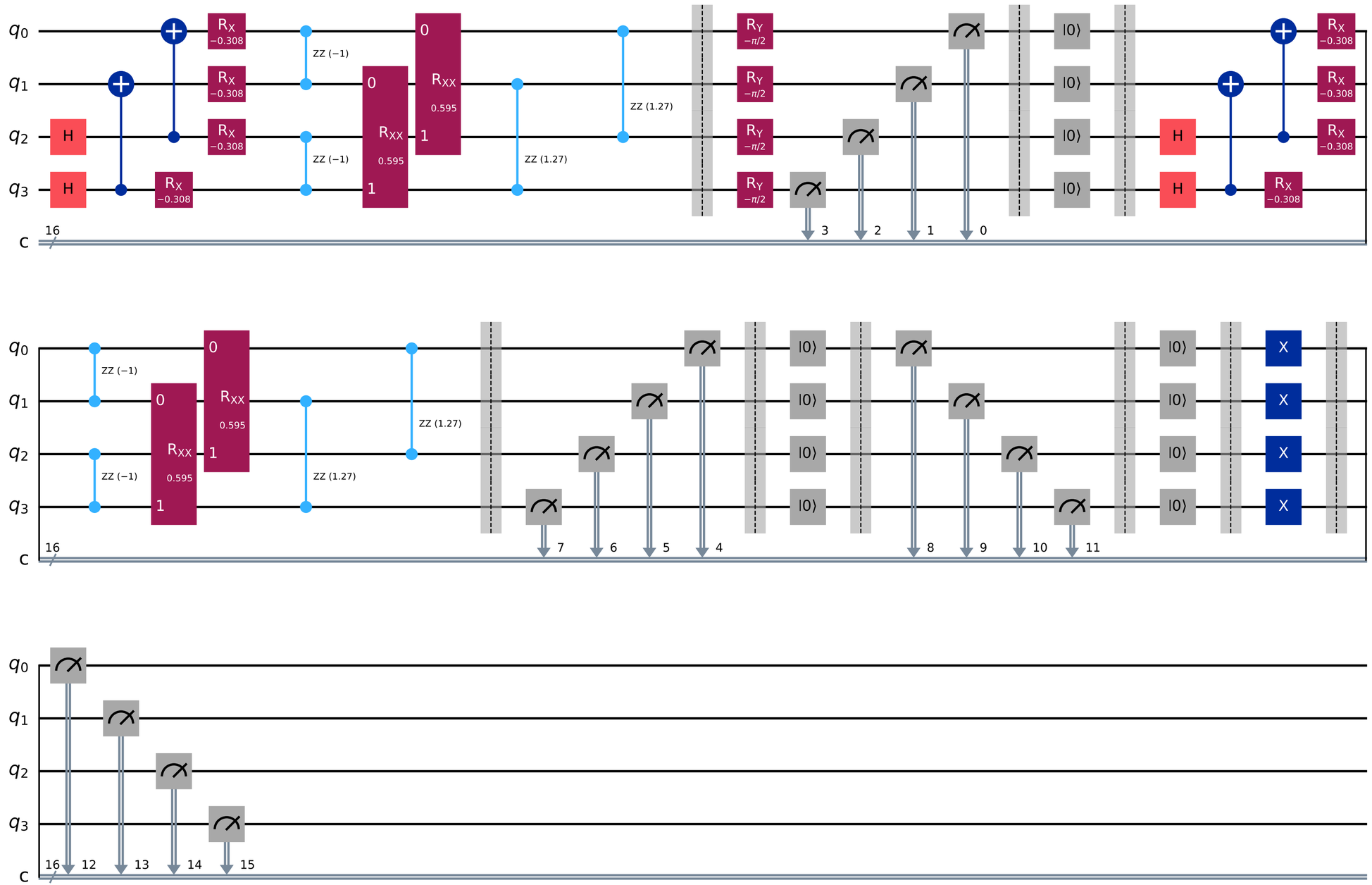

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.169
R_X −0.169
R_X −0.169
R_X −0.169
ZZ (0.321)
ZZ (0.321)
0
R_{XX}
0.714
1
0
R_{XX}
0.714
1
ZZ (0.628)
ZZ (0.628)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.169
R_X −0.169
R_X −0.169
R_X −0.169
ZZ (0.321)
ZZ (0.321)
0
R_{XX}
0.714
1
0
R_{XX}
0.714
1
ZZ (0.628)
ZZ (0.628)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

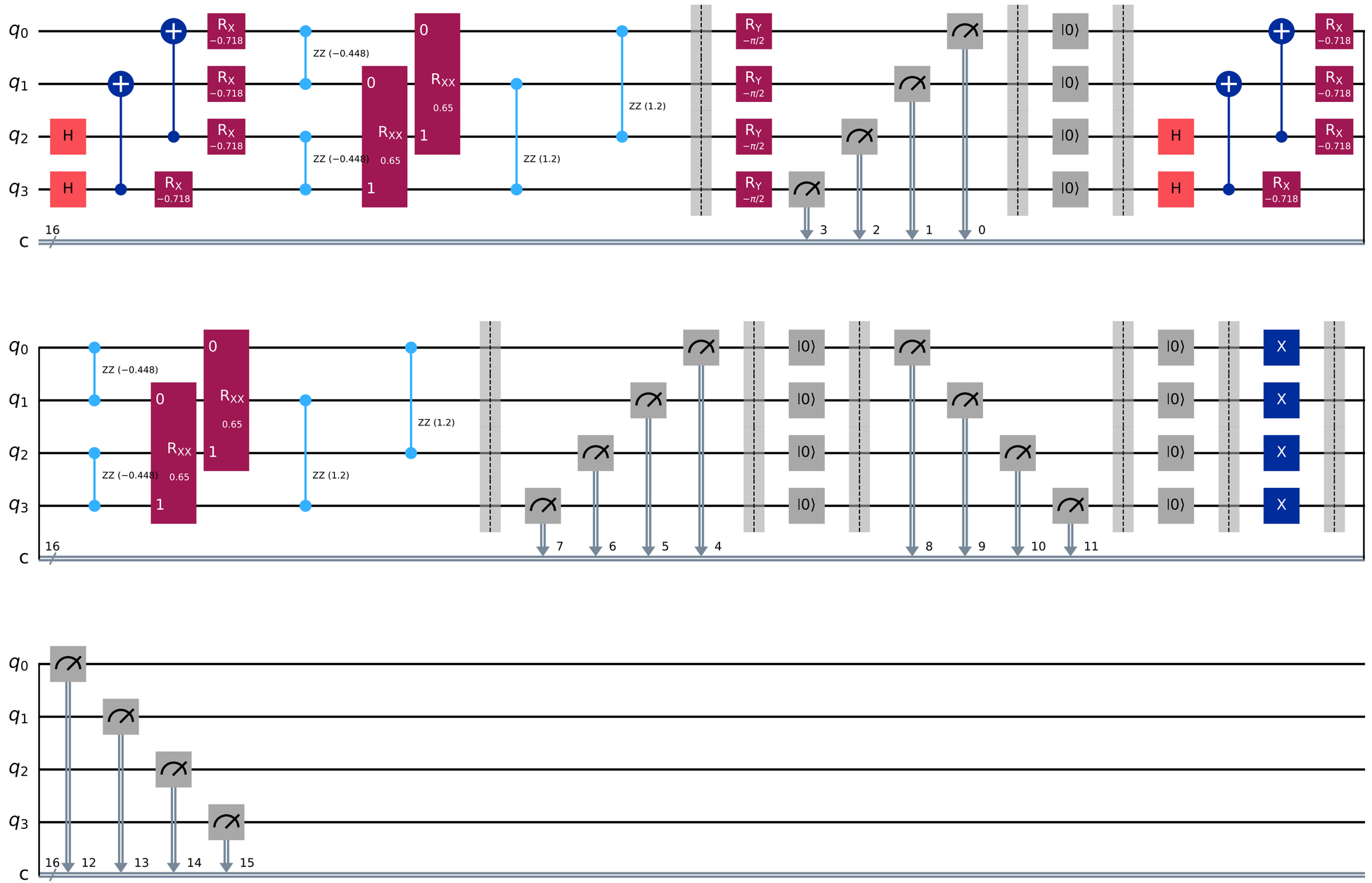

q0
q1
q2
q3
c
16
H
H
RX −0.169
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 1.21
1
0
RXX 1.21
1
ZZ (0.628)
ZZ (0.628)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.169
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 1.21
1
ZZ (0.628)
ZZ (0.628)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

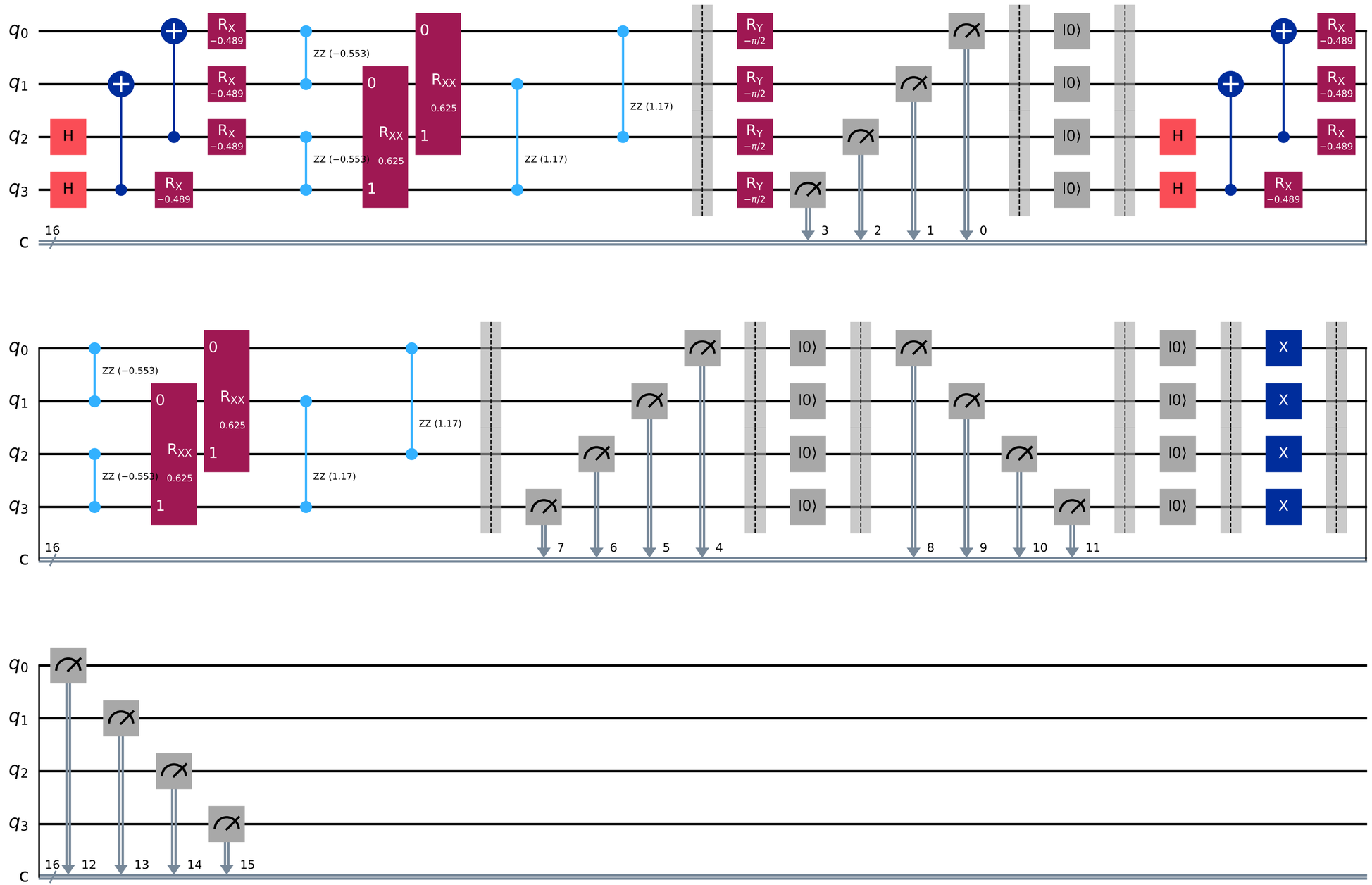

q0
q1
q2
q3
c
16
H
H
RX −0.169
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (1.13)
ZZ (1.13)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (1.13)
ZZ (1.13)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
16
12
13
14
15

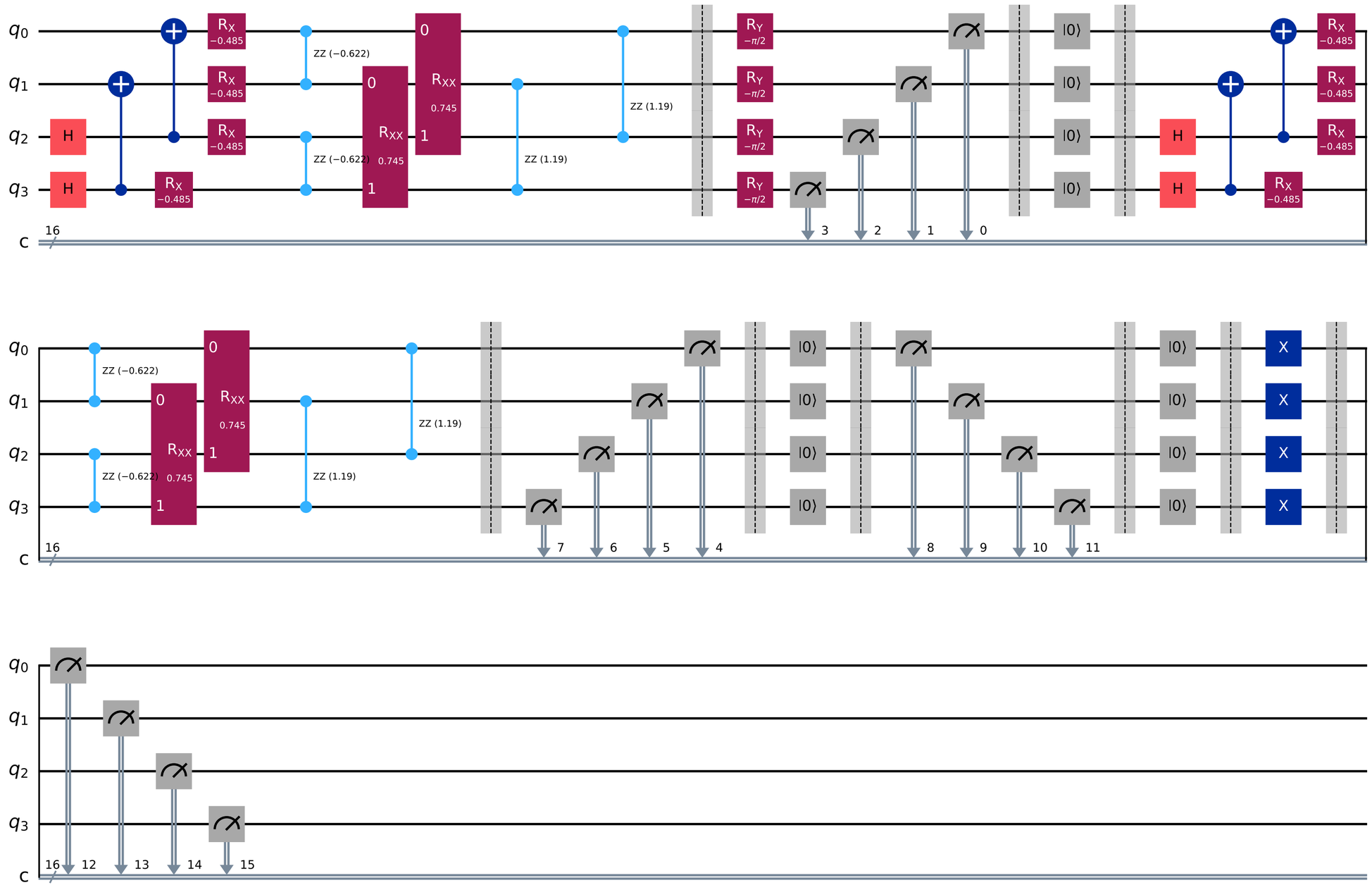

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.495
R_X −0.495
R_X −0.495
R_X −0.495
ZZ (−0.553)
ZZ (−0.553)
R_{XX} 0.675
ZZ (1.17)
ZZ (1.17)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.495
R_X −0.495
R_X −0.495
R_X −0.495
ZZ (−0.553)
ZZ (−0.553)
R_{XX} 0.675
ZZ (1.17)
ZZ (1.17)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

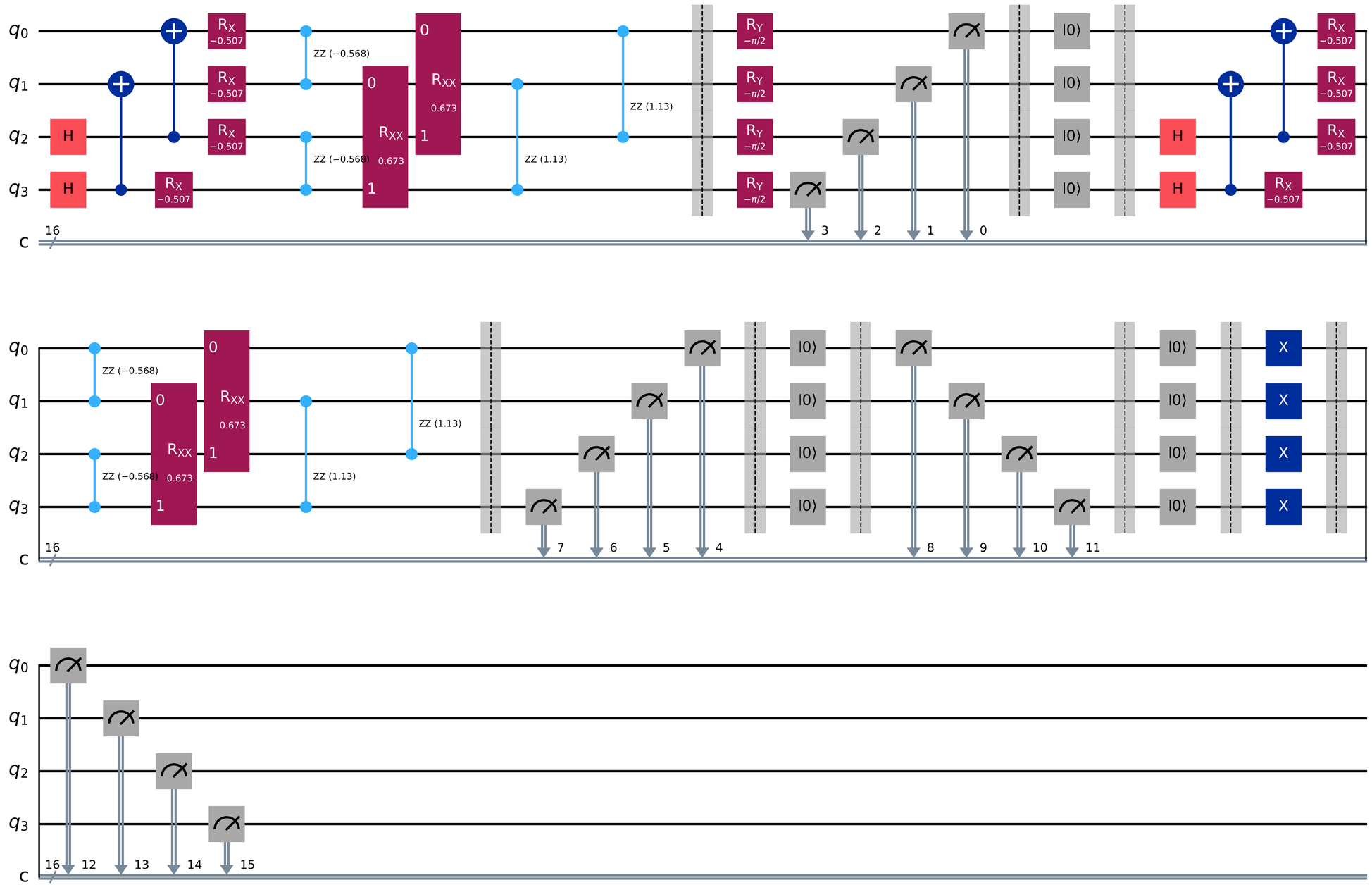

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.946
R_X −0.946
R_X −0.946
R_X −0.946
ZZ (−0.356)
ZZ (−0.356)
R_{XX} 0.62
ZZ (1.24)
ZZ (1.24)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.946
R_X −0.946
R_X −0.946
R_X −0.946
3
2
1
0
ZZ (−0.356)
ZZ (−0.356)
R_{XX} 0.62
ZZ (1.24)
ZZ (1.24)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15

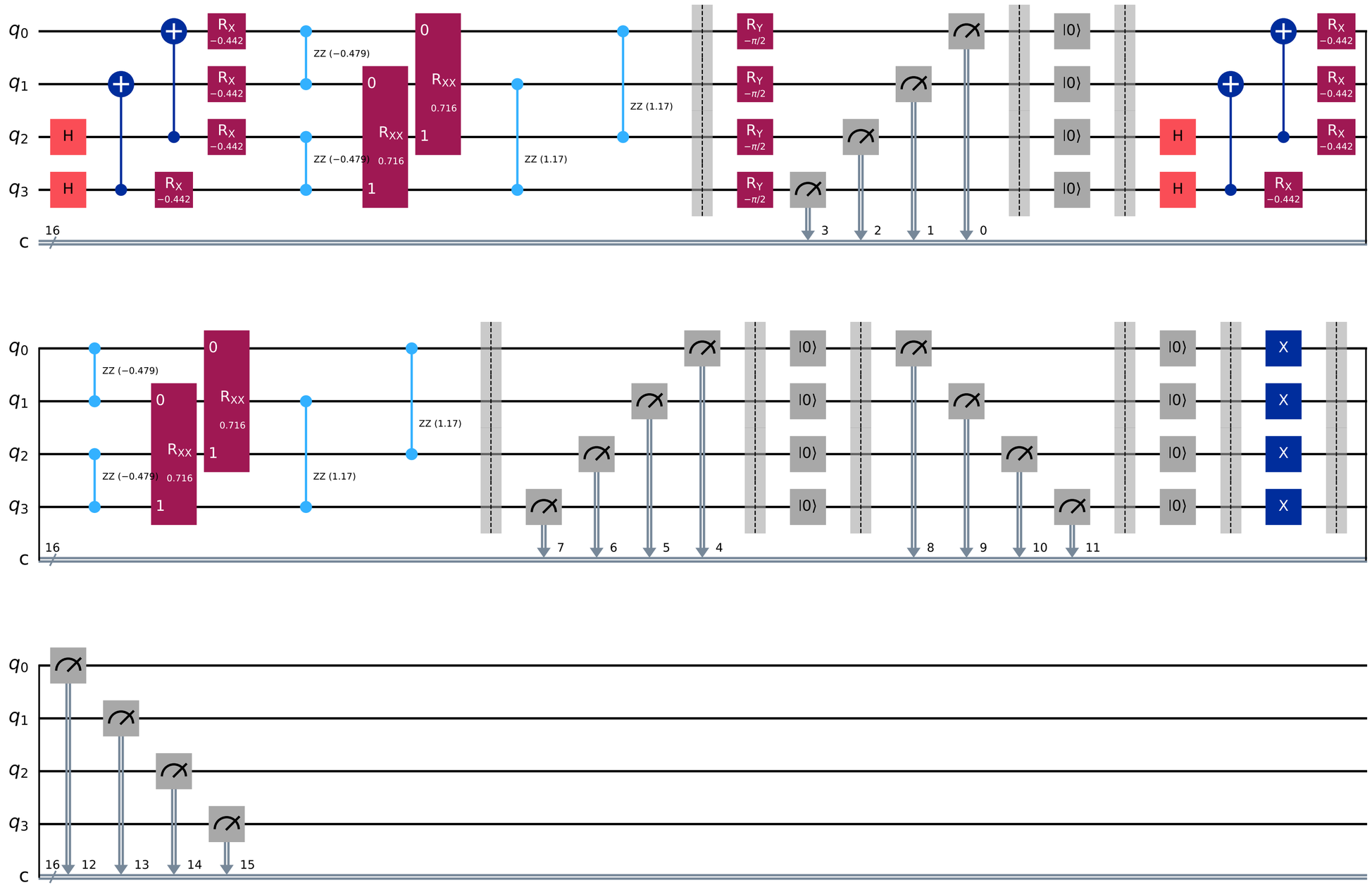

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.308
R_X −0.308
R_X −0.308
R_X −0.308
ZZ (−1)
ZZ (−1)
R_{XX} 0.595
ZZ (1.27)
ZZ (1.27)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.308
R_X −0.308
R_X −0.308
R_X −0.308
ZZ (−1)
ZZ (−1)
R_{XX} 0.595
ZZ (1.27)
ZZ (1.27)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15

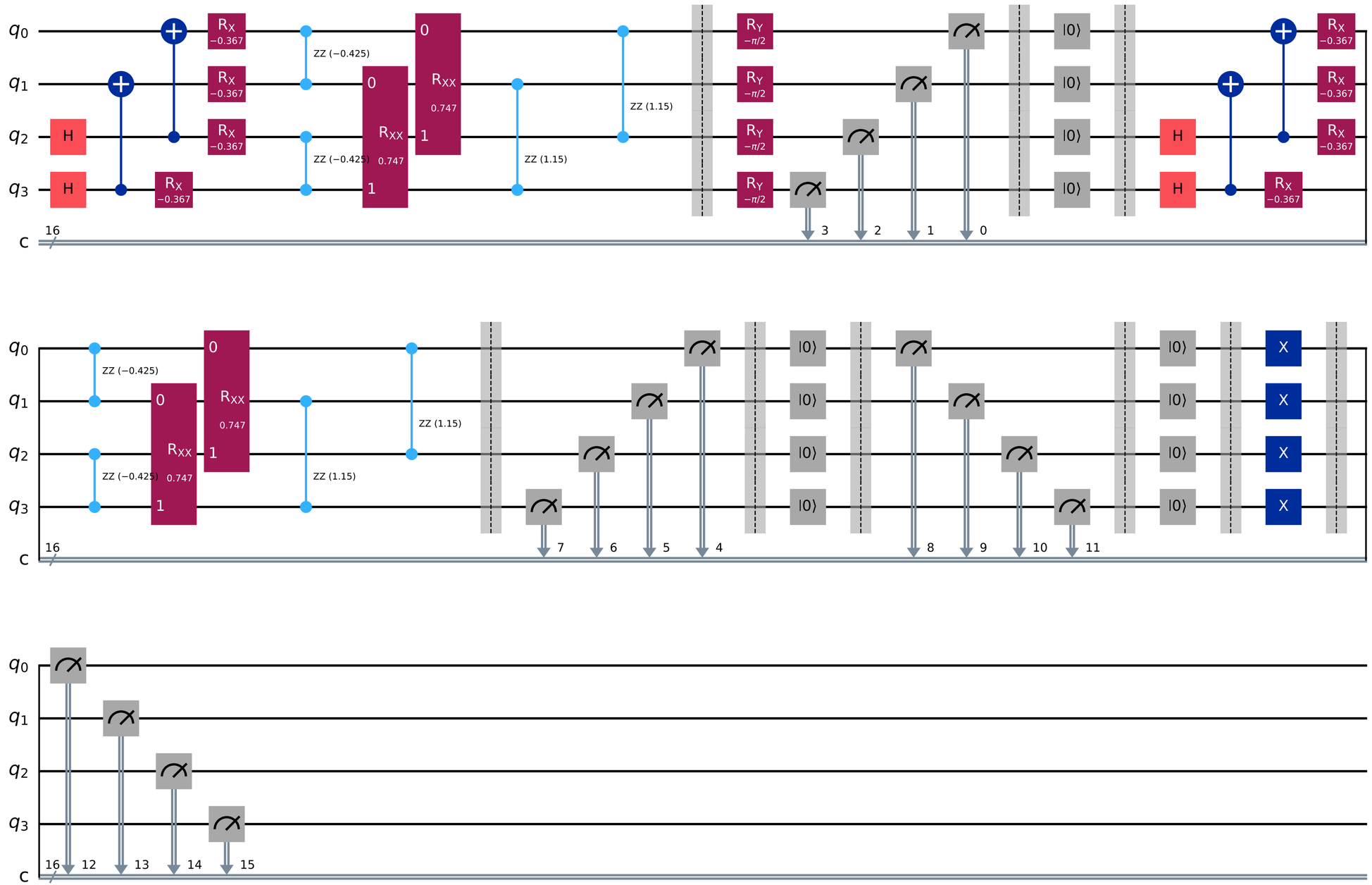

q0
q1
q2
q3
c
16
H
H
RX −0.718
RX −0.718
RX −0.718
RX −0.718
ZZ (−0.448)
ZZ (−0.448)
0
RXX 0.65
1
0
RXX 0.65
1
ZZ (1.2)
ZZ (1.2)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.718
RX −0.718
RX −0.718
RX −0.718
ZZ (−0.448)
ZZ (−0.448)
0
RXX 0.65
1
0
RXX 0.65
1
ZZ (1.2)
ZZ (1.2)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

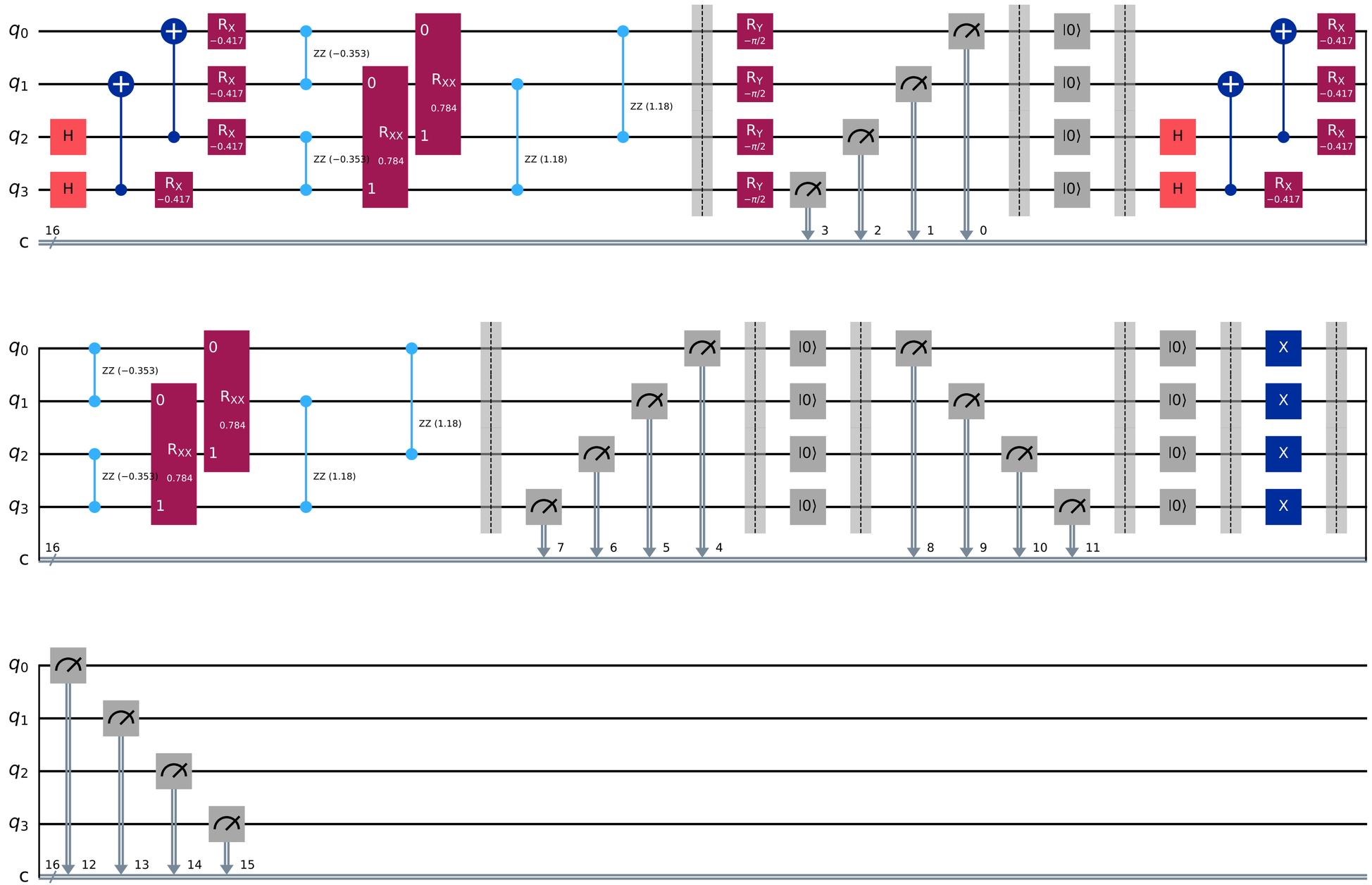

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.489
R_X −0.489
R_X −0.489
R_X −0.489
ZZ (−0.553)
ZZ (−0.553)
R_{XX} 0.625
ZZ (1.17)
ZZ (1.17)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.489
R_X −0.489
R_X −0.489
R_X −0.489
ZZ (−0.553)
ZZ (−0.553)
R_{XX} 0.625
ZZ (1.17)
ZZ (1.17)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15

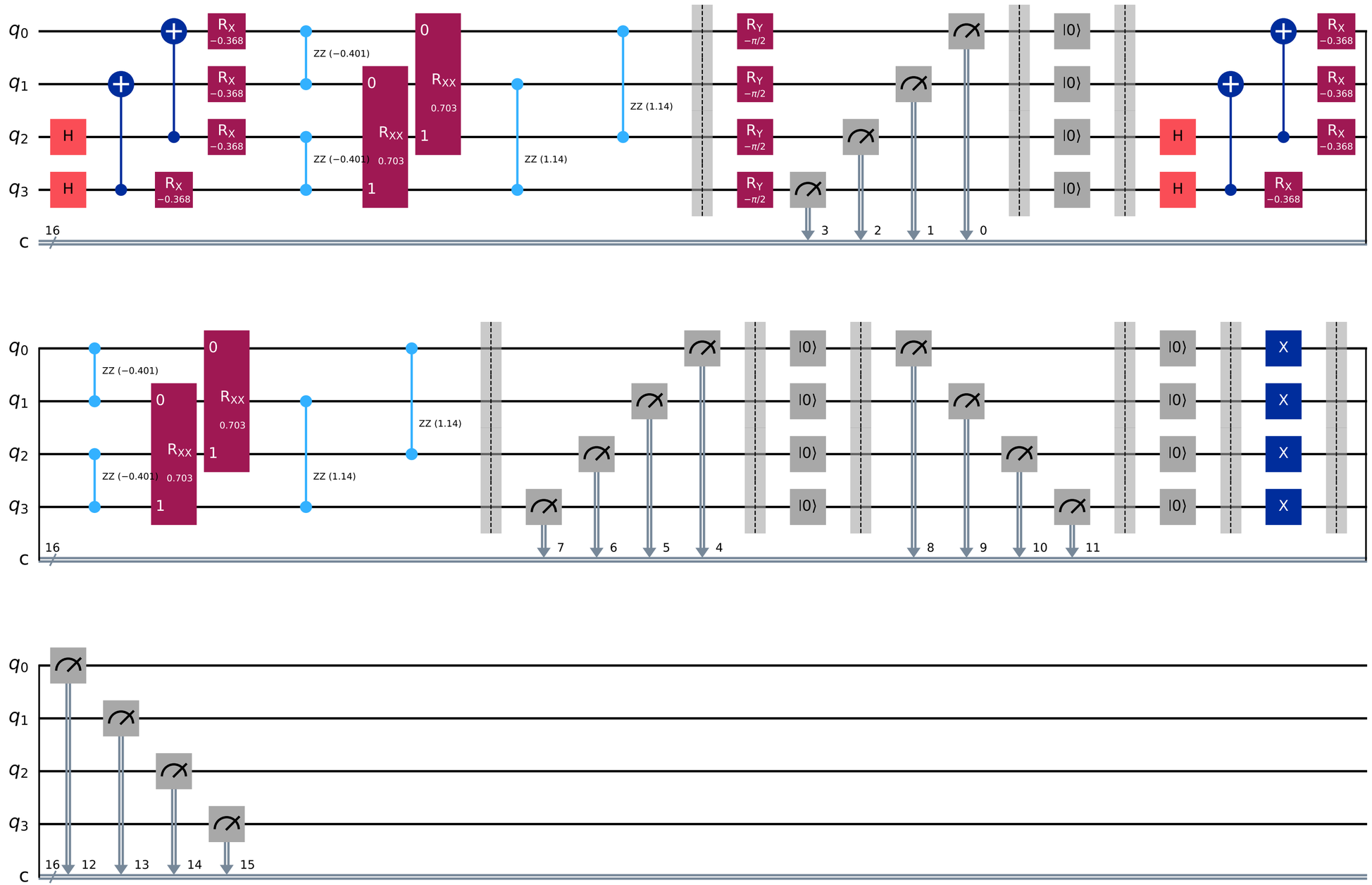

q0
q1
q2
q3
c
16
H
H
RX −0.485
RX −0.485
RX −0.485
RX −0.485
ZZ (−0.622)
ZZ (−0.622)
0
RXX 0.745
1
ZZ (1.19)
ZZ (1.19)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.485
RX −0.485
RX −0.485
ZZ (−0.622)
ZZ (−0.622)
0
RXX 0.745
1
ZZ (1.19)
ZZ (1.19)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
12
13
14
15

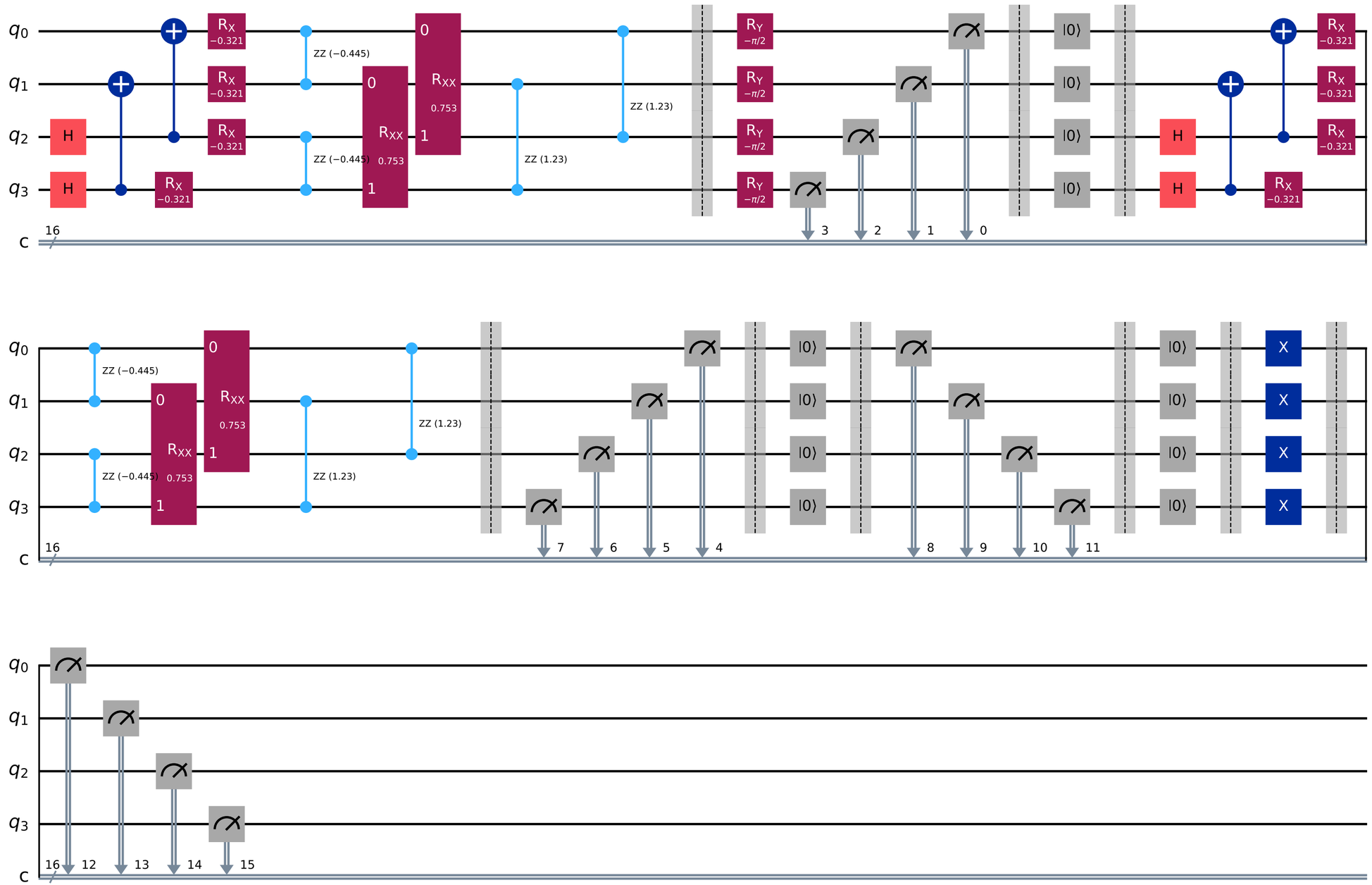

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.507
R_X −0.507
R_X −0.507
R_X −0.507
ZZ (−0.568)
ZZ (−0.568)
0
R_{XX} 0.673
1
ZZ (1.13)
ZZ (1.13)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.507
R_X −0.507
R_X −0.507
R_X −0.507
ZZ (−0.568)
ZZ (−0.568)
0
R_{XX} 0.673
1
ZZ (1.13)
ZZ (1.13)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

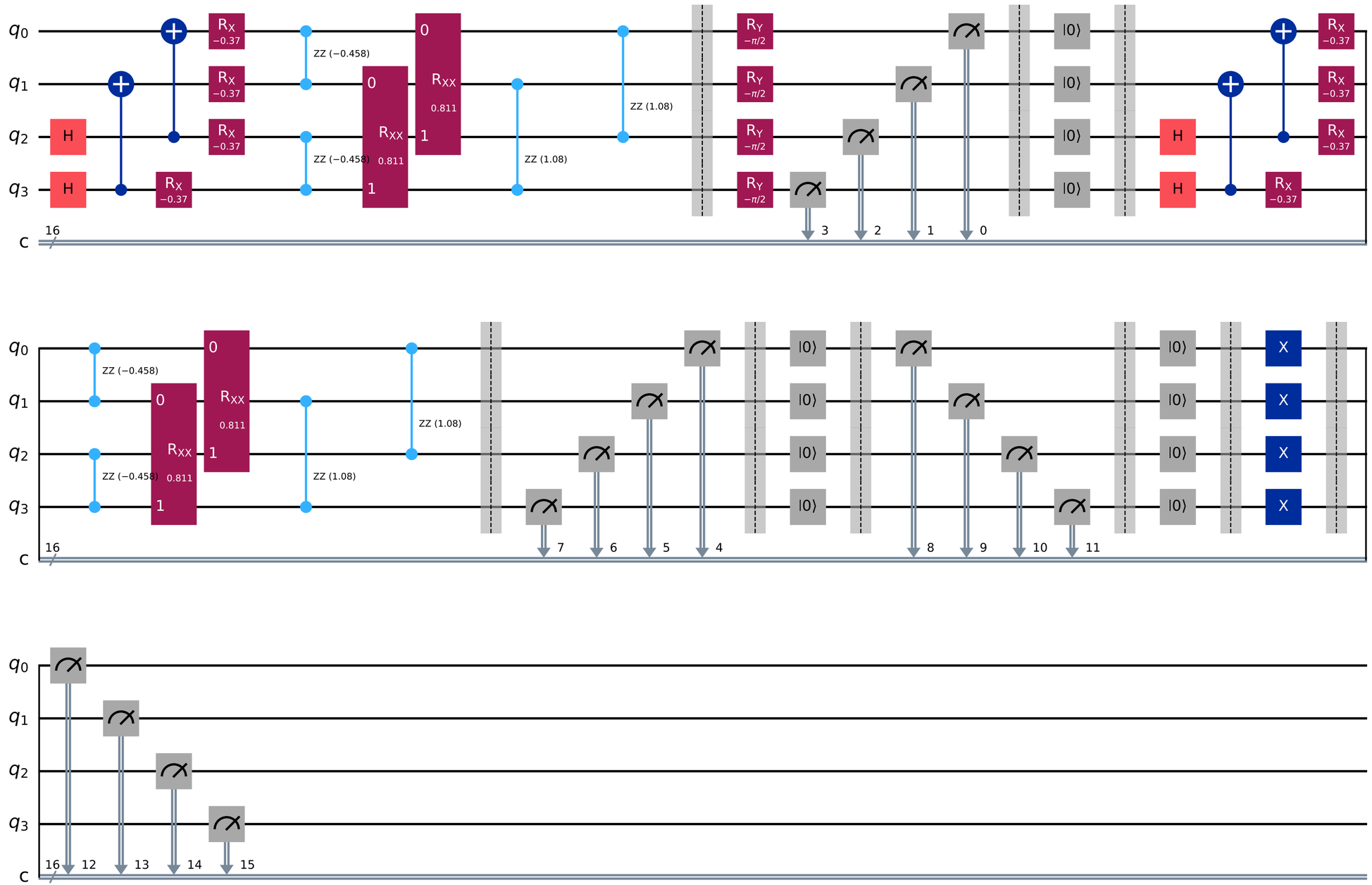

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.442
R_X −0.442
R_X −0.442
R_X −0.442
ZZ (−0.479)
ZZ (−0.479)
0
R_{XX} 0.716
1
0
R_{XX} 0.716
1
ZZ (1.17)
ZZ (1.17)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.442
R_X −0.442
R_X −0.442
R_X −0.442
ZZ (−0.479)
ZZ (−0.479)
0
R_{XX} 0.716
1
0
R_{XX} 0.716
1
ZZ (1.17)
ZZ (1.17)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

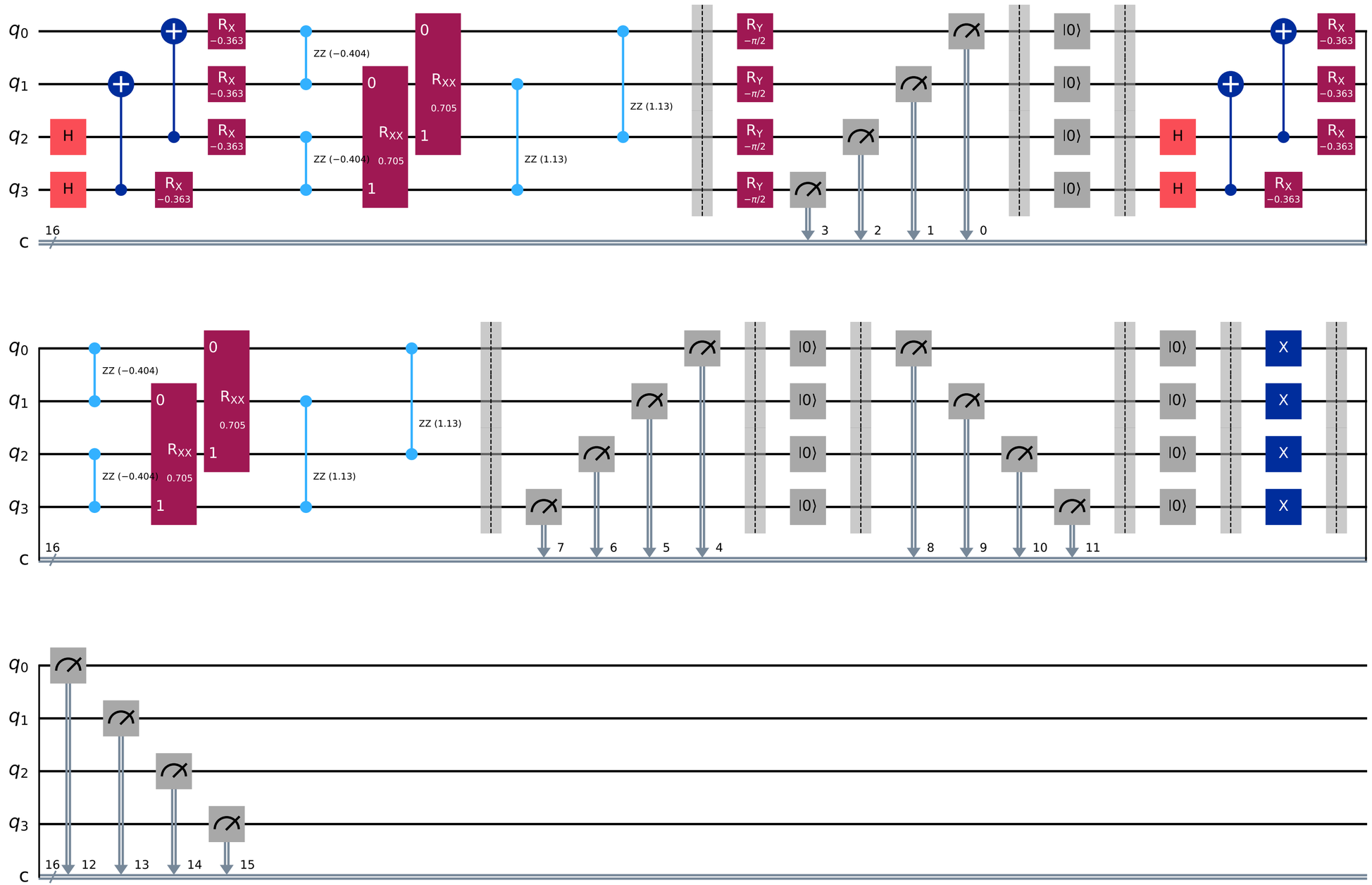

q0
q1
q2
q3
c
16
H
H
RX −0.367
RX −0.367
RX −0.367
RX −0.367
ZZ (−0.425)
ZZ (−0.425)
0
RXX 0.747
1
ZZ (1.15)
ZZ (1.15)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.367
RX −0.367
RX −0.367
RX −0.367
ZZ (−0.425)
ZZ (−0.425)
0
RXX 0.747
1
ZZ (1.15)
ZZ (1.15)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

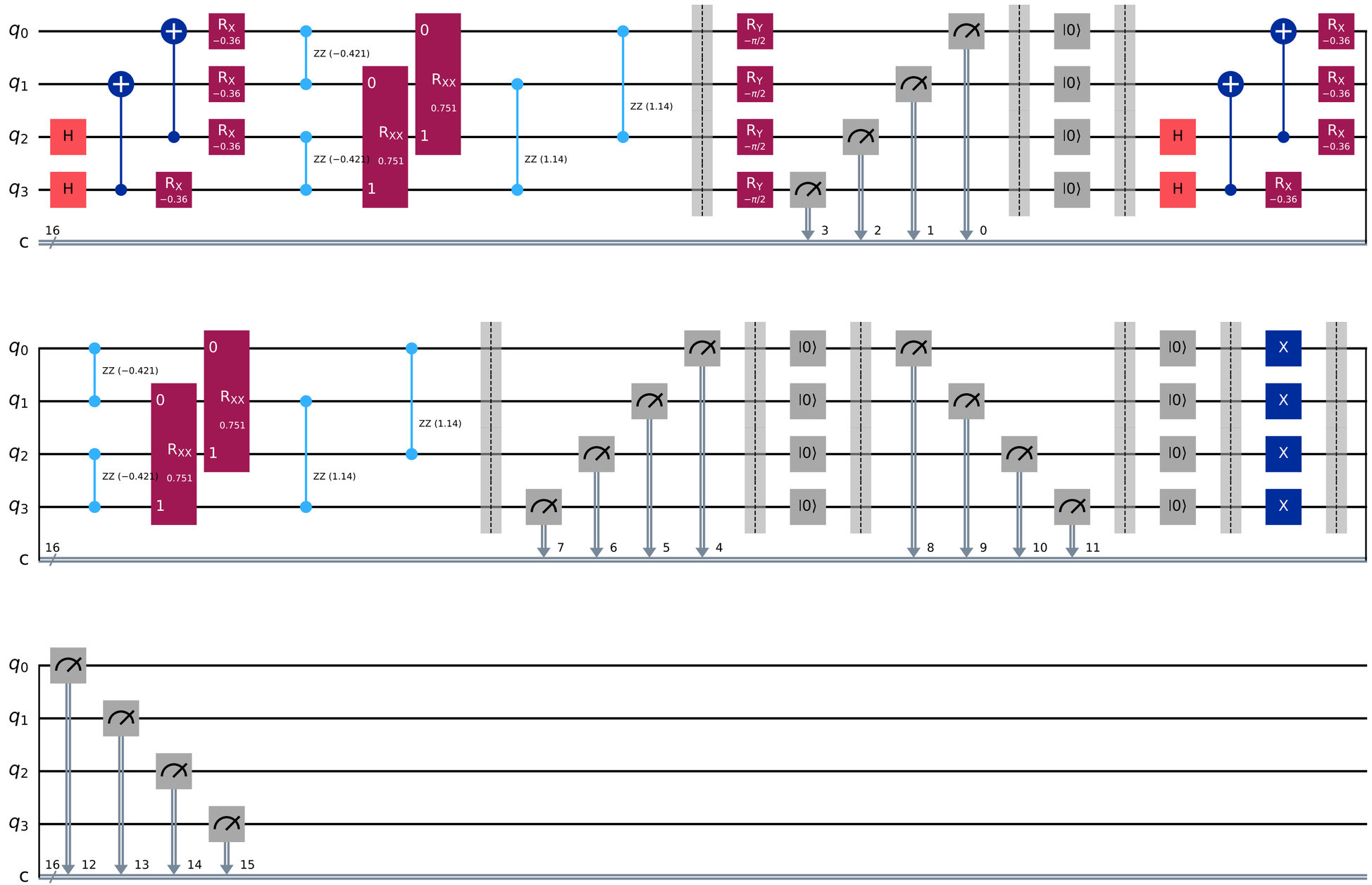

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.417
R_X −0.417
R_X −0.417
R_X −0.417
ZZ (−0.353)
ZZ (−0.353)
0
R_{XX}
0.784
1
0
R_{XX}
0.784
1
ZZ (1.18)
ZZ (1.18)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.417
R_X −0.417
R_X −0.417
R_X −0.417
ZZ (−0.353)
ZZ (−0.353)
0
R_{XX}
0.784
1
0
R_{XX}
0.784
1
ZZ (1.18)
ZZ (1.18)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

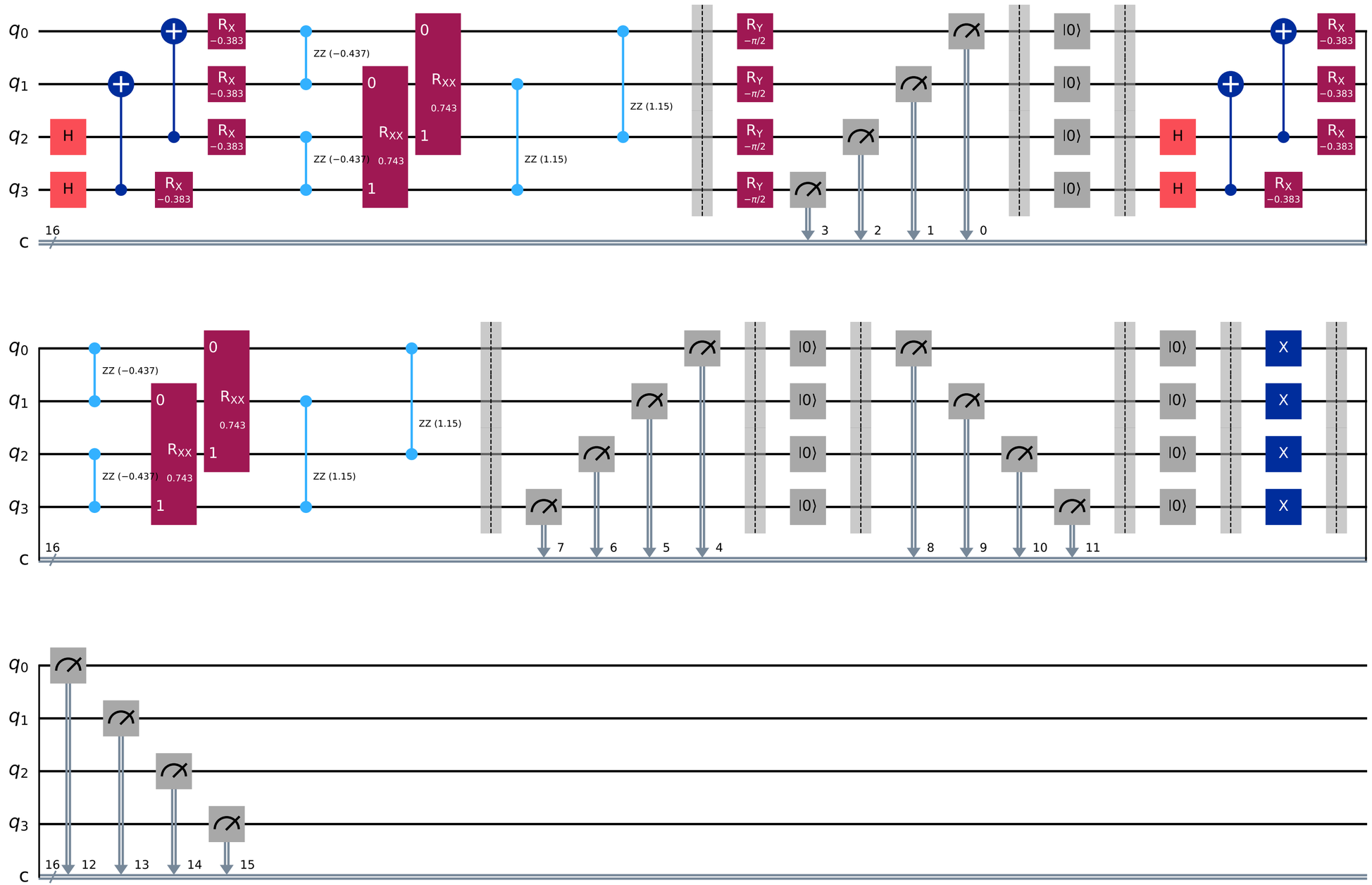

q0
q1
q2
q3
c
16
H
H
RX −0.368
RX −0.368
RX −0.368
RX −0.368
ZZ (−0.401)
ZZ (−0.401)
0
RXX 0.703
1
ZZ (1.14)
ZZ (1.14)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.368
RX −0.368
RX −0.368
RX −0.368
ZZ (−0.401)
ZZ (−0.401)
0
RXX 0.703
1
ZZ (1.14)
ZZ (1.14)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

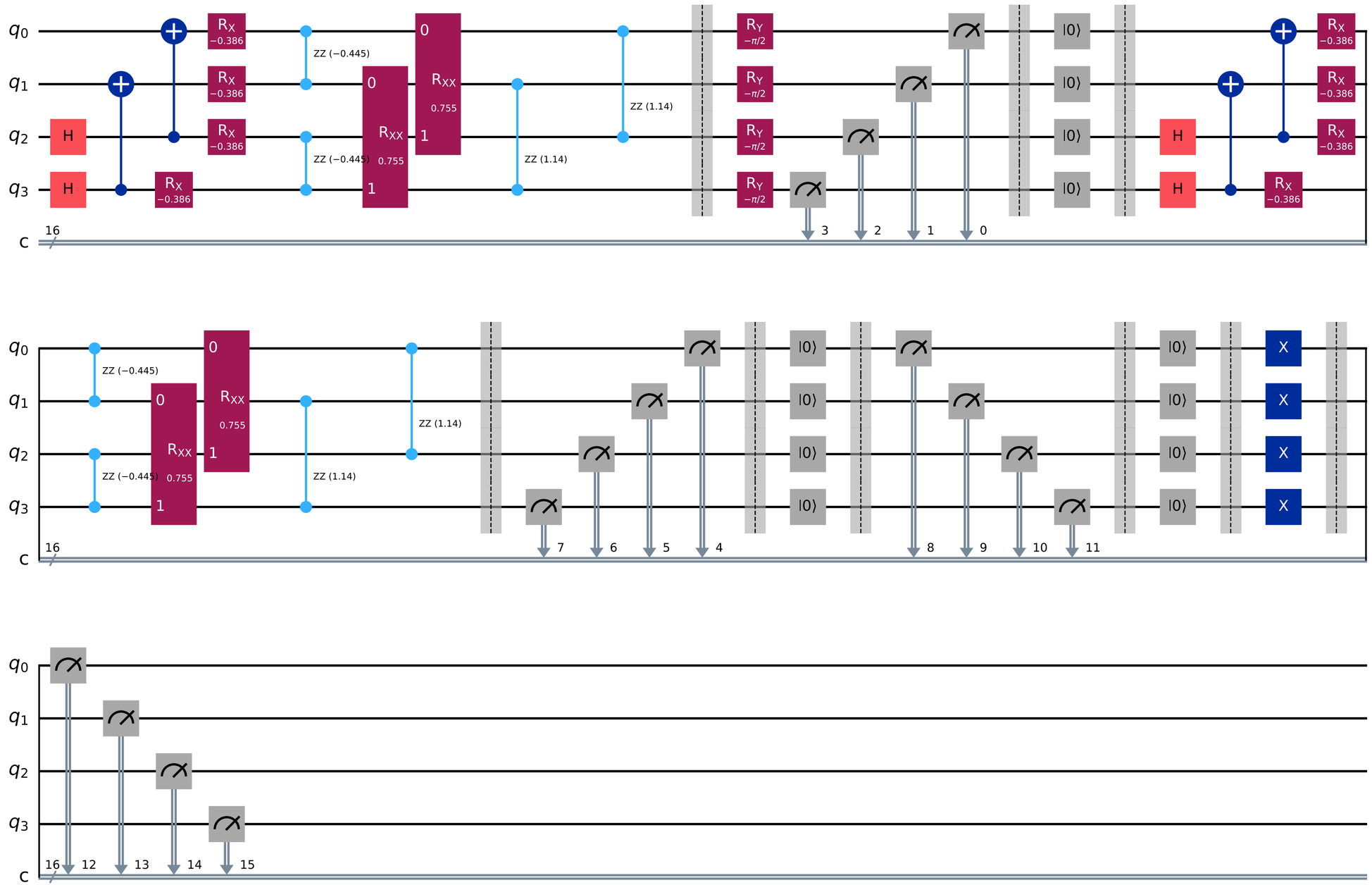

q0
q1
q2
q3
c
16
H
H
RX −0.321
RX −0.321
RX −0.321
RX −0.321
ZZ (−0.445)
ZZ (−0.445)
0
RXX 0.753
1
0
RXX 0.753
1
ZZ (1.23)
ZZ (1.23)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.321
RX −0.321
RX −0.321
RX −0.321
ZZ (−0.445)
ZZ (−0.445)
0
RXX 0.753
1
0
RXX 0.753
1
ZZ (1.23)
ZZ (1.23)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

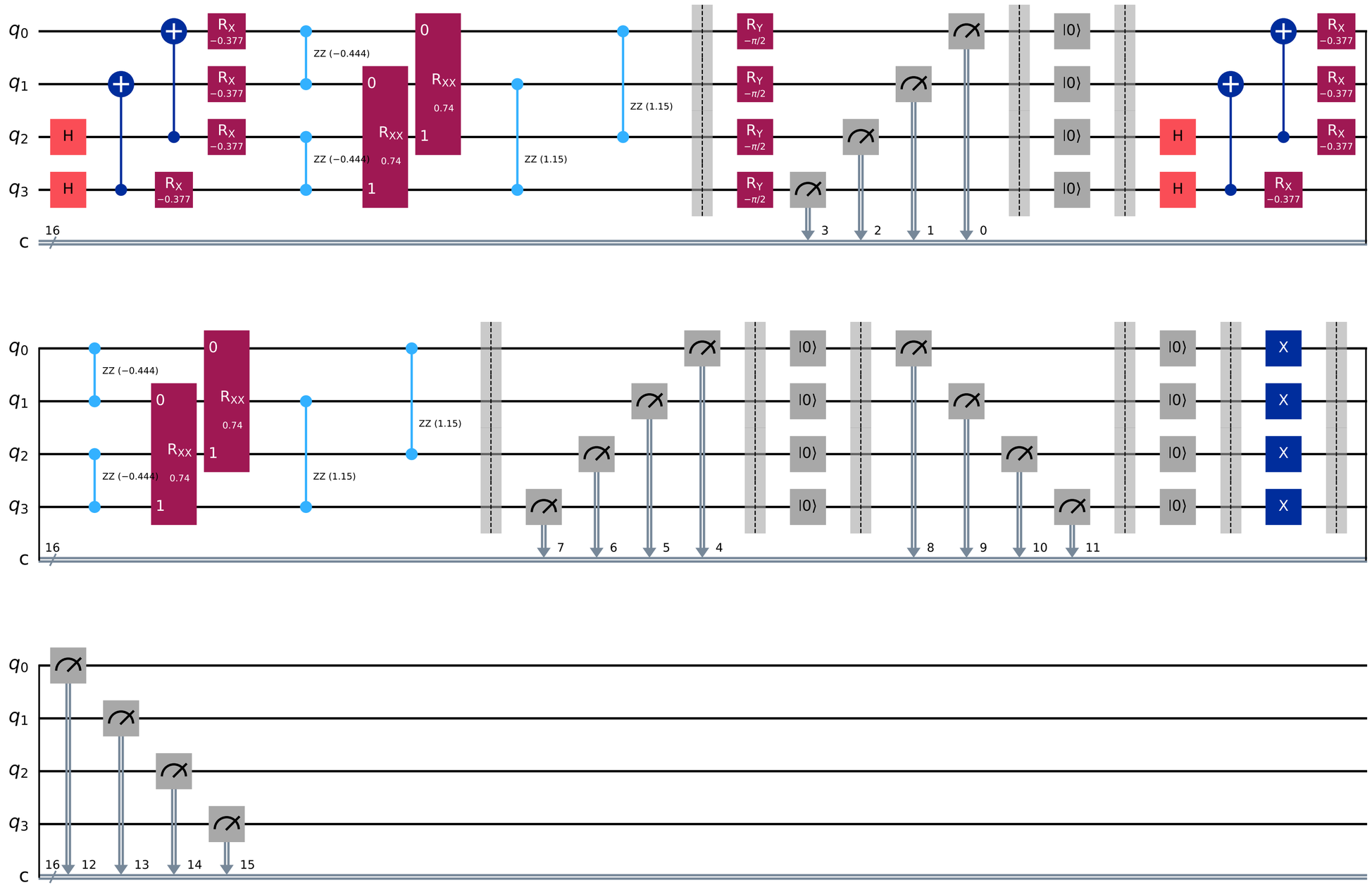

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.37
R_X −0.37
R_X −0.37
R_X −0.37
ZZ (−0.458)
ZZ (−0.458)
0
R_{XX} 0.811
1
ZZ (1.08)
ZZ (1.08)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.37
R_X −0.37
R_X −0.37
R_X −0.37
ZZ (−0.458)
ZZ (−0.458)
0
R_{XX} 0.811
1
ZZ (1.08)
ZZ (1.08)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

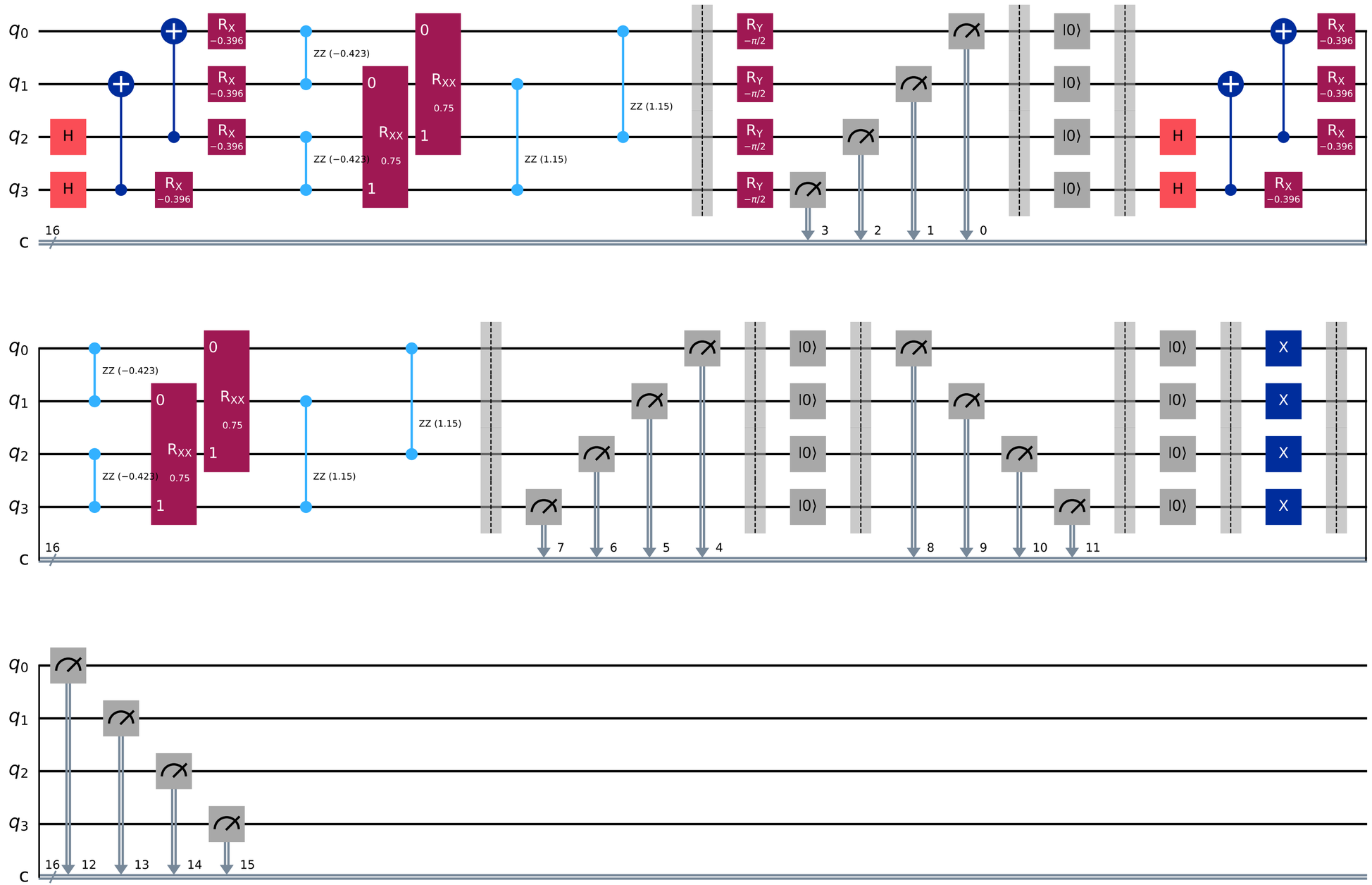

q0
q1
q2
q3
c
16
H
H
RX −0.363
RX −0.363
RX −0.363
RX −0.363
ZZ (−0.404)
ZZ (−0.404)
0
RXX 0.705
1
ZZ (1.13)
ZZ (1.13)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.363
RX −0.363
RX −0.363
RX −0.363
ZZ (−0.404)
ZZ (−0.404)
0
RXX 0.705
1
ZZ (1.13)
ZZ (1.13)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

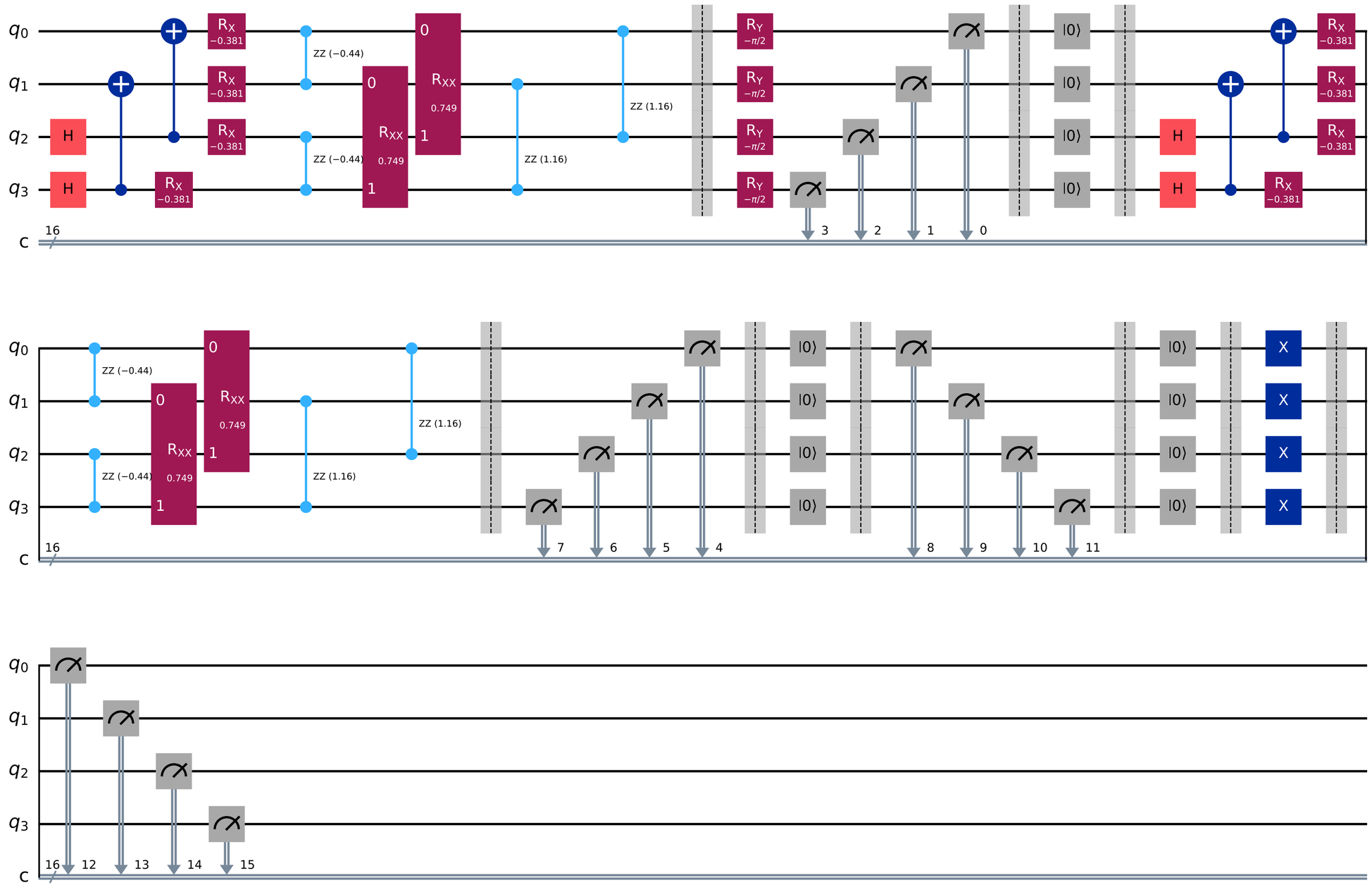

q0
q1
q2
q3
c
16
H
H
RX −0.36
RX −0.36
RX −0.36
RX −0.36
ZZ (−0.421)
ZZ (−0.421)
0
RXX 0.751
1
0
RXX 0.751
1
ZZ (1.14)
ZZ (1.14)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.36
RX −0.36
RX −0.36
RX −0.36
ZZ (−0.421)
ZZ (−0.421)
0
RXX 0.751
1
ZZ (1.14)
ZZ (1.14)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

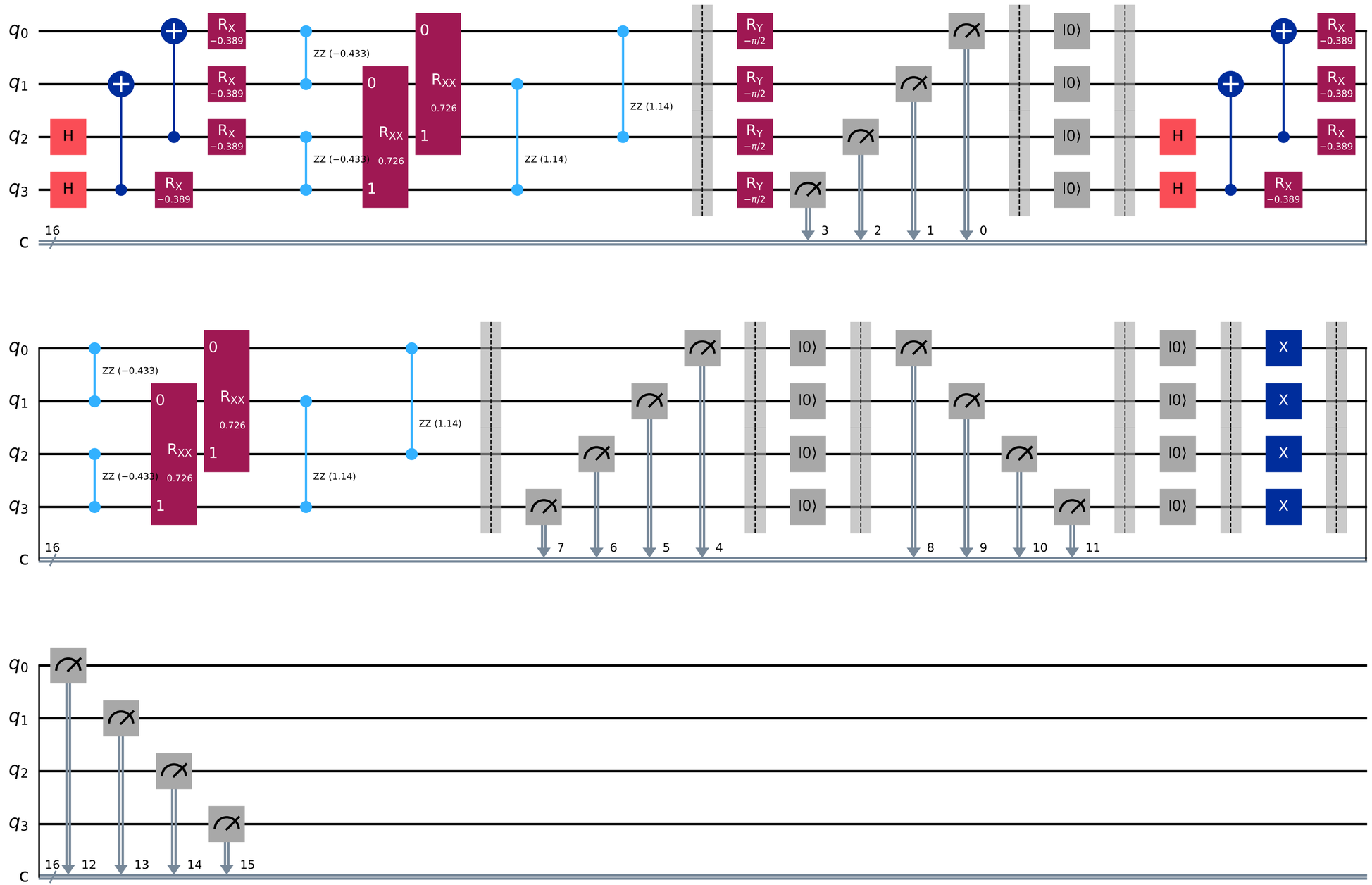

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.383
R_X −0.383
R_X −0.383
R_X −0.383
ZZ (−0.437)
ZZ (−0.437)
R_{XX} 0.743
ZZ (1.15)
ZZ (1.15)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.383
R_X −0.383
R_X −0.383
R_X −0.383
ZZ (−0.437)
ZZ (−0.437)
R_{XX} 0.743
ZZ (1.15)
ZZ (1.15)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.386
RX −0.386
RX −0.386
RX −0.386
ZZ (−0.445)
ZZ (−0.445)
0
RXX 0.755
1
0
RXX 0.755
1
ZZ (1.14)
ZZ (1.14)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.386
RX −0.386
RX −0.386
RX −0.386
ZZ (−0.445)
ZZ (−0.445)
0
RXX 0.755
1
0
RXX 0.755
1
ZZ (1.14)
ZZ (1.14)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.377
RX −0.377
RX −0.377
RX −0.377
ZZ (−0.444)
ZZ (−0.444)
0
RXX 0.74
1
ZZ (1.15)
ZZ (1.15)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.377
RX −0.377
RX −0.377
RX −0.377
ZZ (−0.444)
ZZ (−0.444)
0
RXX 0.74
1
ZZ (1.15)
ZZ (1.15)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.396
RX −0.396
RX −0.396
RX −0.396
ZZ (−0.423)
ZZ (−0.423)
0
RXX 0.75
1
ZZ (1.15)
ZZ (1.15)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.396
RX −0.396
RX −0.396
ZZ (−0.423)
ZZ (−0.423)
0
RXX 0.75
1
ZZ (1.15)
ZZ (1.15)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.381
R_X −0.381
R_X −0.381
R_X −0.381
ZZ (−0.44)
ZZ (−0.44)
R_{XX} 0.749
ZZ (1.16)
ZZ (1.16)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.381
R_X −0.381
R_X −0.381
R_X −0.381
ZZ (−0.44)
ZZ (−0.44)
R_{XX} 0.749
ZZ (1.16)
ZZ (1.16)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.389
R_X −0.389
R_X −0.389
R_X −0.389
ZZ (−0.433)
ZZ (−0.433)
R_{XX} 0.726
ZZ (1.14)
ZZ (1.14)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.389
R_X −0.389
R_X −0.389
R_X −0.389
3
2
1
0
ZZ (−0.433)
ZZ (−0.433)
R_{XX} 0.726
ZZ (1.14)
ZZ (1.14)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15